



The revised learning process questionnaire: A validation of a Western model of students' study approaches to the South Pacific context using confirmatory factor analysis

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Background. Research evidence seems to suggest that the social and cultural environments influence students' approaches to their study. This social and cultural contention has led to the rethinking and reconceptualization of theories (e.g. Biggs, 1987; Marton & Säljö, 1976) pertaining to student approaches to learning (SAL) in academic settings.

Aims. The present research discusses two separate empirical studies on student learning approaches situated in the South Pacific region with two respective cohorts of secondary students. Study I involved the examination of secondary Pacific Islands students in their learning approaches using a modified version of Biggs' (1987) original Learning Process Questionnaire (LPQ). Study II involved the administration of a revised version of the LPQ (R-LPQ-2F; Kember, Biggs, & Leung, 2004) to another cohort of secondary Pacific Islands students.

Sample. The first sample included 2,150 (1,285 girls, 865 boys) students and the second sample included 2,295 (1,363 girls, 932 boys) students.

Methods. The factor structures of approaches to learning were examined by means of confirmatory factor analysis (CFA) using the LISREL program. Different *a priori* models were hypothesized and tested.

Results. Results of Study I indicated a two-factor structure solution to Biggs' LPQ, supporting Richardson's (1994) theoretical model of learning and emphasized the factors of *Reproducing* and *Meaning*. Study II indicated a hierarchical organization of two main study approaches – *deep* and *surface* – that are structured as higher-order factors and a defined by four first-order factors.

Conclusion. The results from the two studies accentuate the important argument for the rethinking and reconceptualization of learning approaches, as well as for the redevelopment and modification of learning inventories such as the LPQ. They also suggest the importance of situating the theoretical paradigm of learning approaches in a social and cultural environment.

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Research concerning the study approaches of students has made substantial ground in the past three decades. Evidence from a number of instruments such as the Approaches to Student Inventory (ASI) (Entwistle & Ramsden, 1983), the Approaches and Study Skills Inventory for Students (ASSIST) (Entwistle, Tait, & McCune, 2000; Tait, Entwistle, & McCune, 1998), the Learning and Study Strategies Inventory (LASSI) (Weinstein & Palmer, 1990) and the Learning (LPQ) and Study Process Questionnaire (SPQ) (Biggs, 1987) confirms the generality of surface and deep study approaches of secondary and tertiary students. However, recently, an extended interest has surfaced in this area of research by Phan and colleague (Deo & Phan, 2006; Phan & Deo, 2006, 2007). First, the authors contended for the reconceptualization and reformulation of the theoretical models of study approaches (Biggs, 1987, 1993a; Richardson, 1994) and learning questionnaires used to measure these learning approaches. Second, the authors addressed the appropriateness and hence constructed the validity of Biggs' (1987) LPQ and SPQ, in particular, with South Pacific students. The emphasis then, in this regard, is to question whether theoretical conceptions of learning should be situated in social and cultural contexts. Whilst extensive research has focused on learners of different social and cultural contexts (e.g. Ainley, 1993; Akande, 1998; Kember, 2000; Zhang & Bernardo, 2000), findings have emerged as being contradictory in terms of the appropriateness of the theoretical models advocated for learning.

In this article, we present two empirical studies situating Biggs' (1987) theoretical framework of learning and the LPQ in a South Pacific context, in this case the Republic of Fiji. The empirical and theoretical emphasis in our two studies is not on the cross-cultural comparison of findings with other countries *per se*, but whether a Western model of learning could be applied to another social and cultural context. Central to our argument is the notion that historical, social and cultural backgrounds of the Fijian people play a major role in influencing their preferred mode of learning. In this analysis, the theoretical conception of learning (Biggs, 1987; Marton & Säljö, 1976) and previous evidence of research is integrated into our two studies. For our two studies, we used the LPQ (Biggs, 1987) and the revised version of the LPQ (LPQ-R-2F; Kember, Biggs, & Leung, 2004) to explore secondary school students' study approaches. We also employed latent variable approaches (Bollen, 1989; Byrne, 1998; Chou & Bentler, 1995) to help establish findings concerning Fijian students' study approaches.

The Fijian educational context

Fiji gained independence from Great Britain in 1970. Much of the educational innovations and policies in the post independence era were largely based on the recommendations of the 1969 Royal Commission. Fiji borrowed and adopted the education systems of metropolitan powers such as Britain, New Zealand and Australia. Examinations were an integral part of education systems of these countries at that time, and the examination model has since become firmly entrenched in Fiji's education systems (Mangubhai, 1984).

With almost 900 schools in Fiji, about 99% are owned and operated by non-government organizations. All the schools and government owned tertiary institutions come under the ambit of the Fiji Ministry of Education. The Ministry prepares and provides the curriculum, examinations, teachers and grants to schools. Children begin primary education at the age of 6 years. They spend 8 years in primary education and 5 years in secondary, and most practice what can be described as rote learning, resulting in dull and unimaginative learning (Government of Fiji, 2000; Mugler & Landbeck, 1997;

Richardson, Landbeck, & Mugler, 1995; Tavola, 1991). It is noteworthy to mention that secondary school students sit for five sets of examinations over a 13-year period.

Student approaches to learning: A theoretical perspective

The theoretical paradigm of SAL (Marton & Säljö, 1976), involving deep and surface approaches to learning, is based on the premise that students approach their studies for various reasons and these reasons influence the way they go about their learning (Murphy & Tyler, 2005; Watkins & Akande, 1994). In theoretical and practical sense, students' motives for learning affect their choice of strategies for learning. The approach to learning that is adopted by a student reflects the interaction between individual characteristics of a student and the context and content of the task involved.

Extending the theoretical framework of SAL, Ramsden (1979) and Biggs (1987) further added an 'achieving' component to the surface and deep study approaches. The achieving approach, according to Biggs, is based on achieving motivation and involves those strategies which students believe will lead to high marks; for example, 'study skills' techniques (e.g. good organization, speed reading, effective note-taking) and 'cue-conscious' strategies that depend on the learning environment and the extent of teacher involvement (Akande, 1998). The achieving approach is therefore context-dependent and the surface and deep approaches relate to rehearsal and the general cognitive processes of Coding, respectively (Akande, 1998). However, importantly, Biggs' (1987) theoretical conception of study approaches differs from other theorists in its two components: how students approach a task (strategy) and why they want to approach it in the first place (motive).

The development and research of the original LPQ

The Learning Process Questionnaire (LPQ) measuring approaches to learning has been used extensively in studies of different social and cultural contexts. The LPQ containing 36 items (Biggs, 1987) was developed and interpreted within the generic framework of SAL theory (Biggs, 1993a). Higher-order factor analysis of the original items resulted in three factors that comprised two kinds of items, those corresponding to a motive and those corresponding to a congruent strategy (Kember *et al.*, 2004). The three motives and three congruent strategies were identified as surface, deep and achieving (see Table 1). The LPQ then yields three approach scores, surface, deep and achieving, respectively, and a component motive and strategy score for each approach. The LPQ elicits students' responses to study approaches that attest to the framework of a presage-process-product (3P) model (Biggs, 1987, 1993a, 1993b).

A review of the literature pertaining to the LPQ indicates two main areas: an examination of the psychometric properties of the LPQ by means of confirmatory factor analysis (CFA) and exploratory factor analysis (EFA), and the study of student learning behaviours in secondary schools using the LPQ. The factor structure of the LPQ and the study approaches that students adopt has formed the basis for extensive cross-cultural research in different countries and cultures (e.g. Bernardo, 2003; Cano, 2005a; Eklund-Myrskog & Claes-Göran, 1999; Kember *et al.*, 2004; Peng & Bettens, 2002; Sachs & Gao, 2000; Watkins & Akande, 1994; Watkins & MBoya, 1997; Wong, Lin, & Watkins, 1996). An analysis of these empirical studies indicates anomalous results amongst secondary students of different countries and cultures. In particular, the inconsistency obtained for non-Western countries attests to the need for

Table 1. Description and sample items for learning process questionnaire

Scale	Description	Sample items
<i>Surface approach</i>		
Motive	Main aim to gain qualifications at minimum allowable standard	Whether I like it or not, I can see that studying is for me a good way to get a well-paid or secure job
Strategy	Reproduce bare essentials using rote learning	I tend to study only what is set; I do not do anything extra
<i>Deep approach</i>		
Motive	Interest in subject and its related areas	I find that my schoolwork can give me a feeling of real satisfaction
Strategy	Understand what is to be learned through detailed inter-relating ideas and reading	While I am studying, I try to think of how useful the material that I am learning would be in real life
<i>Achieving approach</i>		
Motive	To obtain highest possible grades	I really want to do better than anyone else in my school work
Strategy	Organize and design to achieve high marks by being a 'model' student	I try to work solidly throughout the term and review regularly when the examinations are close

additional research to clarify these ambiguous results. The ambiguity of existing research evidence also indicates the important argument concerning the emerging view and possible reconceptualization of students' study approaches, as well as the social and cultural implications that the LPQ addresses.

Factorial analysis of the LPQ has been subjected to CFA in a number of studies. The work of Kember and colleagues in Hong Kong and other countries using CFA indicates two major study approaches: surface and deep. Kember and Leung (1998) tested a number of *a priori* first- and second-order factor models and found the two-factor model as the best fitted model. The two major factors, which are indicative of students' study approaches, and labelled as 'reproducing' and 'meaning', lend support to the conclusion that Richardson (1994) drew, namely that 'there is little unambiguous support for any *separate*, "strategic" approach to academic assessment of the sort that was originally postulated by Ramsden (1979), based upon an "achieving" orientation towards studying' (p. 463). Similarly, Wong *et al.*'s (1996) research has also added credence to Richardson's conclusion concerning the two major factors of reproducing and meaning. In a structural equation analysis, Wong *et al.* tested six models on 10 samples of primary and secondary school LPQ data from Nigeria, Zimbabwe, Malaysia, China, Hong Kong and Canada. Results showed two-factor structures as having better model fit than three-factor models. In a more recent study, Kember *et al.* (2004) using a revised version of the LPQ (LPQ-R-2F), which contained modified and new items derived from the original LPQ (Biggs, 1987), found that secondary school students' study approaches could be interpreted in a hierarchical manner. The hierarchical structure has two second-order factors – deep and surface factors – with four subscales relating to two motive and two strategy factors. The overall hierarchical structure still retains Biggs' (1987) original conceptualization of the motive and strategy components of an approach to learning. Analysis of this study indicates the reformulation of the LPQ as well as the reconceptualization of the theoretical framework of study approaches. For example, in his latter work, Biggs (1999) has contended the abandonment of the achieving approach.

Other CFA studies have also shown support for a two-factor model as well as arguing the need to readdress the inventory's properties. For instance, Cano (2005a, 2005b) found with Spanish and Australian high school students that a two-factor offered the adequate model fit to the data. In contrast, Sachs and Gao's (2000) study of Chinese secondary school students highlighted the low reliability coefficients of the LPQ subscales that are often reflected in other studies (e.g. Andrews, Violato, Rabb, & Hollingsworth, 1994; Kember & Leung, 1998). Based on this result, Sachs and Gao suggested the need to revise some LPQ items and add other items so as to improve the inventory's overall psychometric properties.

The analysis of the LPQ has also been subjected to EFA within a cross-cultural context. Results obtained from a number of EFA studies contribute to the contemporary view of the factor structure of the LPQ. Bernardo (2003) found that the LPQ is valid as a two-factor solution for non-low-achieving Filipino students but not for the low achievers. Similar factor analysis results for both deep and surface approach dimensions were obtained with Malaysian secondary school students (Watkins & Ismail, 1994). In a Finnish study involving three secondary form levels, Eklund-Myrskog and Claes-Göran (1999) found a similar two-factor pattern with deep and achieving motivation and strategies loading on one factor, and surface motivation and surface strategy on the other factor. However, the surface motivation variable loaded on both factors and showed a larger loading on factor I than on factor II. Research in Nigeria (Watkins & Akande, 1994) and Black South Africa (Watkins & MBoya, 1997) of the LPQ also showed two different factors, with one factor consisting of surface motivation and strategy and the other consisting of deep and achieving motivation and strategy.

The present research: A need to readdress the learning process questionnaire

A need to readdress and reconceptualize the learning approaches of students and the LPQ has been a source of research interest recently in the literature (Biggs, Kember, & Leung, 2001; Kember *et al.*, 2004; Sachs & Gao, 2000). This emerging interest stems from two main reasons, amongst them being the inconsistent evidence of the conceptions of learning cross-culturally (e.g. Bernardo, 2003; Eklund-Myrskog & Claes-Göran, 1999; Kember & Leung, 1998; Zhang & Bernardo, 2000), and the concern regarding the psychometric properties of the LPQ. Several studies have to date re-examined theoretically and empirically the LPQ and its appropriateness. Kember *et al.* (2004) from their revision of the LPQ (LPQ-R-2F) reconceptualized a two-factor model depicting deep and surface approaches to learning. Their findings based on the revised LPQ, also advocated a hierarchical conceptualization of study approaches where only two study dimensions are emphasized: deep and surface approaches. Sachs and Gao (2000) similarly suggested the need to revise and add new items to the LPQ due to the inventory's moderate psychometric properties. Previous studies (e.g. Bernardo, 2003; Zhang & Bernardo, 2000) have also questioned the applicability of the LPQ with different cultural groups; for example, the LPQ may not be valid for describing the approaches to learning with different cultural groups of students (Hattie & Watkins, 1981; Watkins & Astilla, 1982).

In view of these contemporary issues and limitations, it is important to extend this line of research inquiry. The two empirical studies reported in this article addressed the study approaches of secondary school students in the South Pacific context, in this case the Republic of Fiji. The original LPQ (Biggs, 1987) and the revised version of the LPQ (LPQ-R-2F; Kember *et al.*, 2004) were administered in two studies to examine their applicability as well as explore Biggs' (1987) theoretical conceptions of students' study

approaches. The renewed interest in our two studies then, concerned exploring the appropriateness of the two inventories in a different social and cultural context.

Study I: The LPQ

This study, based on previous research evidence (e.g. Kember & Leung, 1998) investigated students' study approaches in their final year of secondary schooling using Biggs' (1987) LPQ. We employed CFA to investigate the factor structure of the LPQ as there is theoretical grounding to postulate different *a priori* models (Bollen, 1989; Mueller, 1996; Pedhazur, 1997). A discussion of CFA is beyond the scope of this article; however, CFA has been used recently in a number of studies to explore the dimensions of students' study approaches (e.g. Burnett & Dart, 2000; Cano, 2005a; Fox, McManus, & Winder, 2001; Kember *et al.*, 2004; Kember & Leung, 1998; Sachs & Gao, 2000; Wong *et al.*, 1996). Students' responses were then tested with seven first- and second-order factor models, as presented in Figure 1. Kember and Leung and others (Deo & Phan, 2006; Phan & Deo, 2006, 2007) also tested the seven *a priori* models in their studies and found consistent evidence of a two-factor model descriptive of students' study approaches. In our analyses, the first five models were first-order factor models while the last two were second-order factor models.

Model 1A: Two-factor model

This is a two-factor model based on Richardson's (1994) suggestion. This model has two latent factors: 'meaning' (ME) and 'reproducing' (RP). The meaning factor has an orientation towards comprehending the meaning of the materials to be learnt, while the reproducing factor describes an orientation towards being able to reproduce materials for the purpose of academic assessment. The reproduction factor has the surface strategy and surface motive subscales as indicators, and the meaning factor has the remaining four subscales, DM, DS, AM, AS as indicators.

Model 1B: Two-factor model with shared indicators

This model is a refinement of Model 1A in that the reproduction factor also has AS and AM as indicators.

Model 2A: Three-factor model

This model has three latent factors: deep approach (DA), surface approach (SA) and achieving approach (AA). Each of the three factors has two indicators representing the strategy and motive subscales.

Model 2B: Three-factor model with correlation b/w (DA) and (AA)

This is an extension of Model 2A with a covariance included between the deep and achieving approaches. The inclusion of the covariance suggests that deep approach is commonly found in conjunction with the achieving approach (Biggs, 1987).

Model 2C: Three-factor model with correlation b/w (DA) and (AA), and (SA) and (AA)

This model is an extension of Models 2A and 2B with another covariance included the surface and achieving approaches.

Model 3A: Higher-order factor model with factors b/w (DA) and (AA)

This model is based on Biggs' (1987) depiction of a composite higher-order deep-achieving factor (D-A).

Model 3B: Higher-order factor model with factors b/w (DA) and (AA), and (SA) and (AA)

This model is an extension of Model 3A with another higher-order factor included for surface-achieving (S-A).

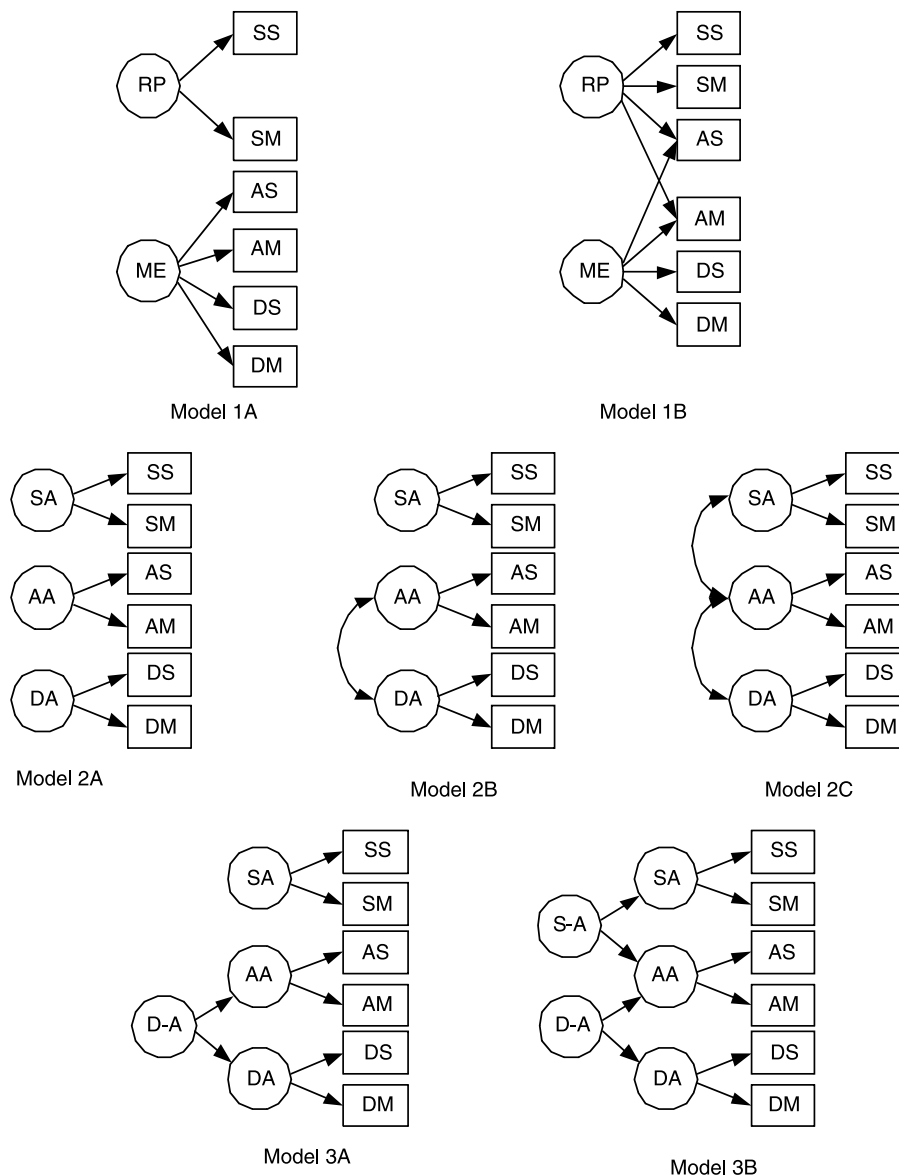


Figure 1. Seven *a priori* structural models of the LPQ (Model 1A to Model 3B). Keys: SS, surface strategy; SM, surface motive; DS, deep surface; DM, deep motive; AS, achieving surface; AM, achieving motive; ME, meaning; RP, reproducing.

Sample

A sample of 2,150 (1,285 girls, 865 boys) secondary school students was drawn from 33 secondary schools across the three divisions (Central, Western, Northern) of Fiji Islands. The sample comprised 662 Indigenous Fijians (427 girls, 235 boys), 1,399 Indo Fijians (798 girls, 601 boys) and 89 others (i.e. Europeans, Rotumans, Chinese; 60 girls, 29 boys). The ages of students ranged from 16 to 23 years (mean = 18.12 years). All students attended Form 7, the highest class level for secondary schools in the Republic of Fiji.

Instrument

The LPQ (Biggs, 1987) is a self-report inventory containing 36 statements about student approaches to learning. Students were asked to respond to each statement by indicating how true that statement is of them using a 5-point Likert scale: *never or only rarely true* (1), *true of me half the time* (3), *always or almost always true of me* (5). The 36 items (see Table 1) are grouped into 6 subscales with 6 items (statements) per subscale. The 6 subscales, in accordance with Biggs' (1987) theoretical conception, are: surface motive (SM), surface strategy (SS), deep motive (DM), deep strategy (DS), achieving motive (AM) and achieving strategy (AS).

Data analysis

The SPSS statistical program was used to determine the reliability of the scales as well as to provide descriptive results. Confirmatory factor analysis (CFA) was used in the present study to examine the learning style orientations of secondary students. CFA, in contrast to other simplistic methodologies (e.g. EFA), is advantageous in its ability to test *a priori* models about latent variables and calculating all of the parameters in the model simultaneously with the overall model fit provided (Bollen, 1989; Farrell, 1994; Kline, 1998; Pedhazur, 1997). The analyses of the seven *a priori* models hypothesized were conducted using the statistical LISREL (Joreskog & Sorbom, 2000) mainframe computer program. Various goodness-of-fit indexes (e.g. CFI, NNFI) were used as indicators representative of a well-fitted model. LISREL, in this case, involved testing the fit between a sample covariance matrix and a matrix reproduced when parameters are constrained to match a theoretical model. Covariance matrices were computed from input raw data and analysed subsequently. The maximum likelihood method (ML) was chosen as the estimation procedure as it is shown to perform reasonably well with multivariate normally distributed data (e.g. Chou & Bentler, 1995).

Results of Study I

The reliability coefficient for the overall inventory was .85, whereas the coefficient α values for the LPQ approaches yielded the following: .65 for surface approach, .76 for deep approach and .77 for achieving approach. For the subscales, Cronbach's α values yielded .47 for surface motive, .58 for deep motive, .53 for achieving motive, .53 for surface strategy, .67 for deep strategy and .78 for achieving strategy. There has been some debate in the literature concerning alpha values and their levels of acceptance (Kember *et al.*, 2004). Whilst some researchers used the cut-off levels of .70, others (e.g. Schmitt, 1996) argue that alpha values as low as .50 would not seriously attenuate validity coefficients. In citing previous work (Lord & Novick, 1968; Schmitt, 1996), Kember *et al.* suggest that there are two factors besides reliability which impact upon the values for Cronbach alpha. First, alpha values are a function of the number of items in a scale, and secondly, alpha can be an underestimate of reliability in the presence of multidimensionality.

For the confirmatory factor analyses, the goodness-of-fit indexes are presented in Table 2. The various goodness-of-fit indexes (e.g. CFI, NNFI) indicated the two-factor, shared indicators model (Model 1B) as providing the best model fit whereas, in comparison, the original three-factor model (Model 2A) provided the worst fit. Similar to Kember and Leung's (1998) study, our results also showed the higher-order factor model (Model 3B) as having a good model fit. The correlated three-factor model

(Model 2C) also showed a moderate model fit. However, a comparison of the goodness-of-fit indexes indicated overall, based on our subjective judgment, that the best fit for the LPQ data was Model 1B (e.g. $CFI = 0.91$, $NNFI = 0.80$). The selection and acceptance between first- and second-factor models is quite subjective from a researcher's point of view. In most cases, the various GFI values between first- and second-order factor models would be near identical with no significant differences in GFIs values in a practical sense. Accordingly, researchers would, based on their own subjective judgments and criteria, choose between first- and second-order factor models of a 'best fitted model' to the empirical data (e.g. Marsh & Hocevar, 1985). Figure 2 shows the standardized solutions for Model 1B with all paths statistically significant at the 0.05 level.

Table 2. Fit indexes of first and second-order factor models

Model	Description	<i>df</i>	χ^2	CFI	NNFI
1A	Two-factor model	9	568.89	0.85	0.75
1B	Two-factor model with shared indicators	7	350.37	0.91	0.80
2A	Three-factor model	9	2020.94	0.47	0.11
2B	Three-factor model with correlation b/w (DA) and (AA)	8	568.60	0.85	0.72
2C	Three-factor model with correlation b/w (DA) and (AA), and (SA) and (AA)	7	472.39	0.88	0.74
3A	Higher-order factor model with factors b/w (DA) and (AA)	7	1766.73	0.53	0.00
3B	Higher-order factor model with factors b/w (DA) and (AA), and (SA) and (AA)	4	235.20	0.94	0.77

Note. NNFI, non-normed fit index; CFI, comparative fit index.

Summary of Study 1

A number of some important findings emerged from the analysis of the different *a priori* models tested. A comparison of the goodness-of-fit indexes for the seven factor models indicated a rejection of Biggs' (1987) theoretical conception of a three-factor structure of study approaches. Consistent with previous empirical studies (e.g. Cano, 2005a, 2005b; Kember & Leung, 1998; Wong *et al.*, 1996), the finding, showed instead an acceptance of a two-factor shared indicators model. Notably, this finding, in concert with other empirical studies cited earlier, provided support for Richardson's (1994) argument that the achieving/strategic dimension could not be classified as a third category of the learning approach construct. When taken together, there is support for a reconceptualization of the learning approaches model as well as Biggs' (1987) LPQ. This finding also supported earlier EFA studies (e.g. Bernardo, 2003; Eklund-Myrskog & Claes-Göran, 1999; Watkins & Akande, 1994; Zhang & Bernardo, 2000) which emphasized a two-factor model for study approaches.

Study 2: The revised LPQ (LPQ-R-2F)

This study, based on Kember *et al.*'s (2004) recent empirical study, concerned the testing of a revised version of Biggs' (1987) LPQ. Kember *et al.* found the revised LPQ (LPQ-R-2F) from

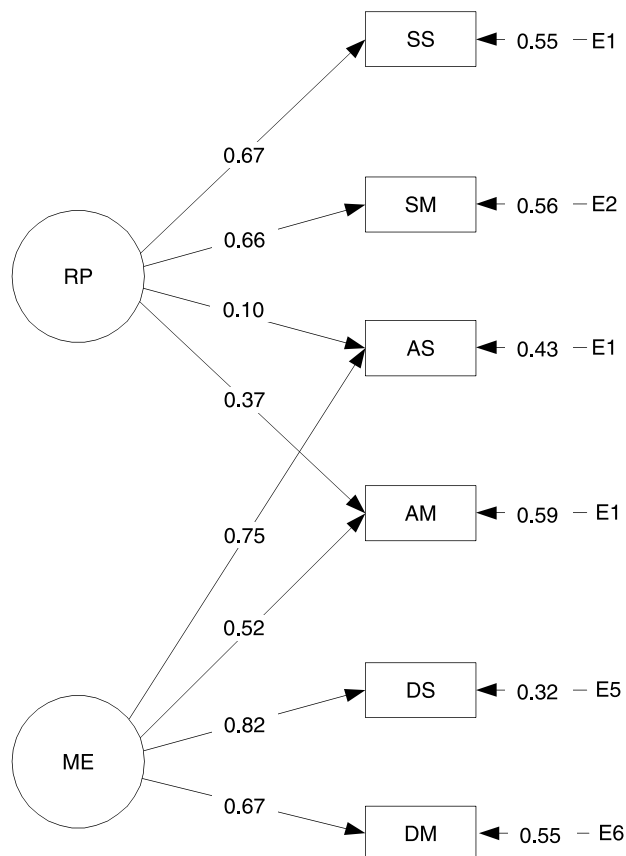


Figure 2. Complete standardized solutions of the model (Model 1B) showing the relationship between the different styles of learning. Keys: variables in circles are latent constructs and variables in squares are observable measures; RP, reproducing; ME, meaning; SS, surface strategy; SM, surface motive; AS, achieving strategy; AM, achieving motivation; DS, deep strategy; DM, deep motive. All paths are statistically significant at the 5% level.

CFA with secondary students from Hong Kong that student study approaches could be conceptualized as a hierarchical structure. Furthermore, the revision of the LPQ from reliability and construct validity measures showed a total of 22 items factoring into two distinct constructs, deep approach and surface approach, and four corresponding subscales – deep motive, deep strategy, surface motive, and surface strategy. The subcomponents of the four subscales, or measured indicators, include intrinsic interest, commitment to work, relating ideas, understanding, fear of failure, aim for qualification, minimizing scope of study and memorization. A discussion of these subcomponents and their correspondences to the four subscales is beyond the scope of this article, however a full explanatory account is presented in Kemmer *et al.* study. The emphasis in this study, instead, focused on examination of the LPQ-R-2F with another cultural group of secondary school students. Central to this study's objective was to establish, empirically, the psychometric property of the LPQ-R-2F, as well as to determine its appropriateness in describing student study approaches compared to Biggs' (1987) original LPQ.

Sample

A sample of 2,295 secondary students (1,363 girls, 932 boys) was drawn from 35 schools across the three divisions of Fiji islands. In nearly all cases, students who participated in Study 1 also took part in Study 2. The total sample consisted of 700 Indigenous Fijians (451 girls, 249 boys), 1483 Indo Fijians (835 girls, 648 boys) and 112 others (77 girls, 35 boys). The time proximity between the two studies was approximately 10 weeks apart. The 10-week duration was not deliberate on our part, but more of a convenience for those involved. Given that these students were in the final year of schooling, time constraint was therefore a pertaining issue to consider.

Instrument

The 22 items of the R-LPQ-2F are grouped into four subscales, deep motive (DM), deep strategy (DS), surface motive (SM) and surface strategy (SS), and eight corresponding subcomponents. Students' responses to each statement were rated on a 5-point Likert scale: *never or only rarely true of me* (1), *true of me about half the time* (3) and *always or almost true of me* (5). The sample items include, for example, 'I find that at times studying makes me feel really happy and satisfied' (intrinsic interest), 'I spend a lot of my free time finding out more about interesting topics which have been discussed in different classes' (commitment to work), 'I try to relate what I have learned in one subject to what I learn in other subjects' (relating ideas), 'I try to relate new material, as I am reading it, to what I already know on that topic' (understanding), 'I am discouraged by a poor mark on a test and worry about how I will do on the next test' (fear of failure), 'Whether I like it or not, I can see that doing well in school is a good way to get a well-paid job' (aim for qualification), 'I see no point in learning material which is not likely to be in the examination' (minimizing scope of study) and 'I learn some things by rote, going over and over them until I know them by heart' (memorization). A complete version of the 22 items may be obtained from Kember *et al.* (2004).

Results of Study 2

Cronbach's alpha values for the two approaches, four subscales and eight components are presented in Table 3. The results in Table 3 indicate the two main scales as having α values above .60. It is noted also that the alpha values for these two subscales as well as the subcomponents are somewhat lower in values when compared with those reported by Kember *et al.* (2004). In this case, only three out of the four subscales have α values exceeding the .50 criterion. However, as explained earlier, the question of Cronbach's alpha values is an issue of much debate.

Table 3. Cronbach's alpha values for the two approaches, four subscales and eight components

Construct	Cronbach
Deep approach	.70 (11 items)
Surface approach	.62 (11 items)
Subscales	
Deep motive	.58 (7 items)
Deep strategy	.54 (4 items)
Surface motive	.42 (4 items)
Surface strategy	.53 (7 items)

The dimensionality of the LPQ-R-2F from CFA included five separate first- and second-factor analyses. Figure 3 shows the initial structure of a two-factor model depicting deep and surface approaches as latent factors. Each latent structure has 11 items as measured indicators. The goodness-of-fit indexes were moderate in values, for example, $CFI = 0.80$, $NNFI = 0.78$, $RMSEA = 0.06$. Comparing with Kember *et al.*'s (2004) results showed similar goodness-of-fit values, indicating that study approaches are better explained by hypothesized hierarchical models than by a unidimensional one.

The factorial structure for deep approach, similar to Kember *et al.*'s (2004) study, is presented in Figure 4. Figure 4 shows the structure for deep approach with four latent constructs, contributed to by the 11 items or measured indicators. The goodness-of-fit values were, for example, $CFI = 0.95$, $NNFI = 0.93$, $RMSEA = 0.05$. It is also noted that the correlations between the four latent constructs were quite high. A correlation value exceeding .90 in this case can raise questions concerning discriminant validity of latent constructs (Byrne, 1998; Phan, 2003). However, the goodness-of-fit indexes suggested a good fit to the data.

Figure 5 shows the fitted model for surface approach with the goodness-of-fit indexes been moderate in values, for example, $CFI = 0.89$, $NNFI = 0.84$ and $RMSEA = 0.06$. Comparing with the previous factorial model of deep approach indicated a poorer model fit, and the latent constructs were positively related but to a lesser extent.

The two factorial models presented in Figures 4 and 5 established the precedence for analysis of a higher-order structure of deep and surface approaches to learning. According to Biggs (1987), each deep and surface study approach consisted of having deep motive and deep strategy components, and likewise surface motive and surface strategy components. Figure 6 shows the higher-order structure relating the eight subcomponents of the approaches identified in Figures 4 and 5. Each motive/strategy factor had two of the components as indicators. Similar to Kember *et al.*'s (2004) findings, deep motive and deep strategy, and surface motive and surface strategy were positively correlated to a large extent. The goodness-of-fit indexes, $GFI = 0.97$, $NNFI = 0.93$, $RMSEA = 0.05$, indicated a good model fit.

The hierarchical model was the final model tested. Figure 7 shows the two highest order latent constructs as deep and surface approaches. The second-order constructs consisted of motive and strategy components of the two approaches. Each of these four latent factors had two subcomponents as indicators. It is a common practice in CFA to

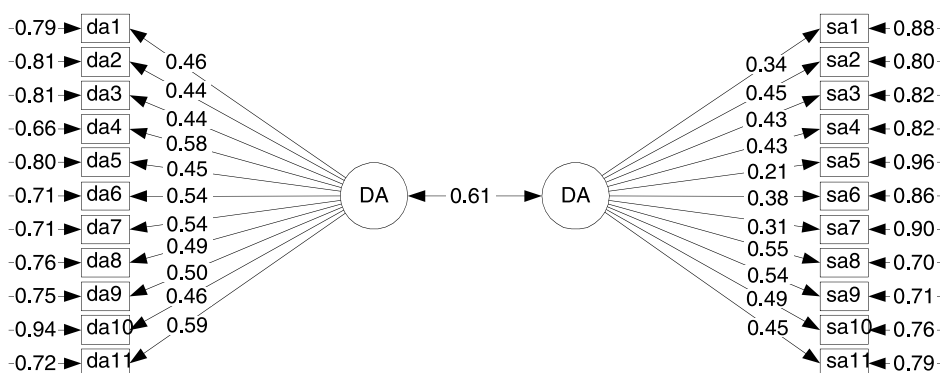


Figure 3. Two-factor structure for deep and surface approach in LPQ-R-2F.

Note: all paths are statistically significant at $p < .05$ and $.01$ levels.

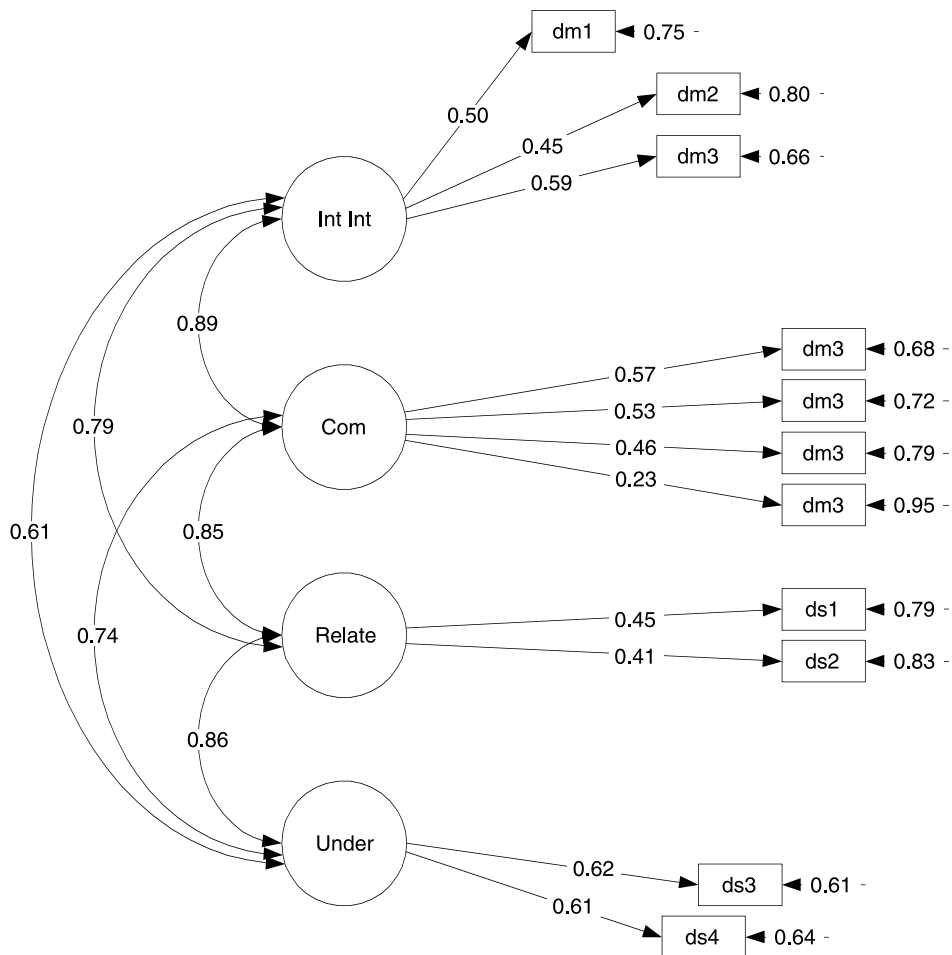


Figure 4. Dimensionality of deep approach in LPQ-R-2F.

Note: Int Int, intrinsic interest; Com, commitment to work; Relate, relating ideas; Under, understanding. All paths are statistically significant at $p < .05$ and $.001$ levels.

specify a value of 1.0 to some of the factor loadings as unidentified models, with the parameters being subject to arbitrariness, often have an infinite number of solutions and cannot be evaluated empirically. Furthermore, it is not appropriate to perform higher-order factor analyses when first-order factors are not sufficiently correlated, statistically. Figure 6 indicated that the four latent factors were statistically correlated.

It is also misleading to use the various goodness-of-fit indexes (e.g. GFI, NNFI) of higher-order factor models to evaluate the extent to which covariances and variances of lower-order factors are accounted for by higher-order factors (Bong, 1997, 2001). Researchers (e.g. Phan, 2003; Vispoel, 1995) have instead suggested comparing the fit of correlated and uncorrelated lower-order factor structures. Comparing the χ^2 and the various goodness-of-fit values, for example $\Delta\chi^2(6, N = 2,295) = 1,376.75, p < .001$, indicated an improvement in fit between a correlated and an uncorrelated model. Finally, the target coefficient (TC) has also been recommended (e.g. Marsh & Hocevar, 1985; Vispoel, 1995) as it reflects the proportion of lower-order factor variances that is

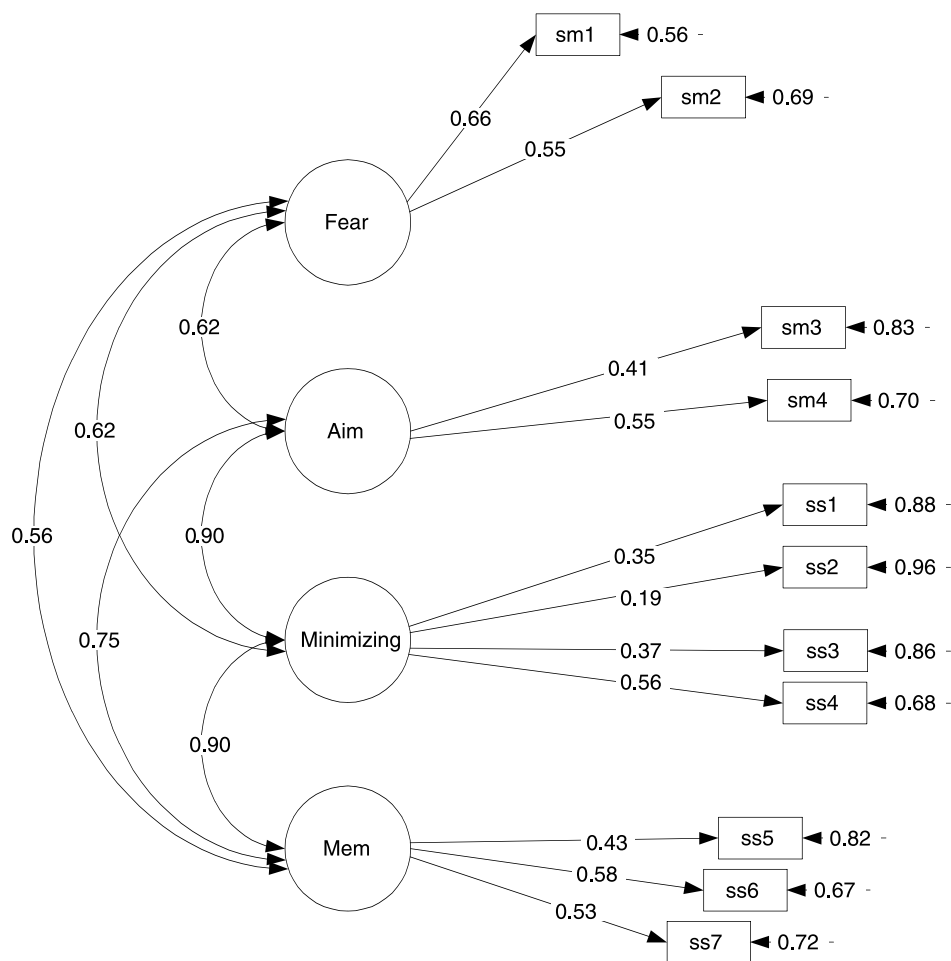


Figure 5. Dimensionality of surface approach in LPQ-R-2F.

Note: Fear, fear of failure; Aim, aim for qualification; Minimizing, minimizing scope of study; Mem, memorizing. All paths are statistically significant at $p < .05$ and $.01$ levels.

accounted for by the higher factors. In this case, the TC for the hierarchical model in Figure 7 was 1.0, indicating that 100% of the covariances among the four first-order factors was accounted for by the two higher-order factors. The goodness-of-fit indexes for the hierarchical model were $GFI = 0.97$, $NNFI = 0.94$ and $RMSEA = 0.05$.

Summary of Study 2

The results of first- and second-order factor analyses indicated moderate support for the revised version of the LPQ in its psychometric properties. An examination of the reliability estimates and the construct validity showed findings that are, to an extent, similar to those reported in the Kember *et al.* (2004) study. The reliability coefficients for the scales and subscales were slightly lower in values when compared with those alpha values reported by Kember *et al.*

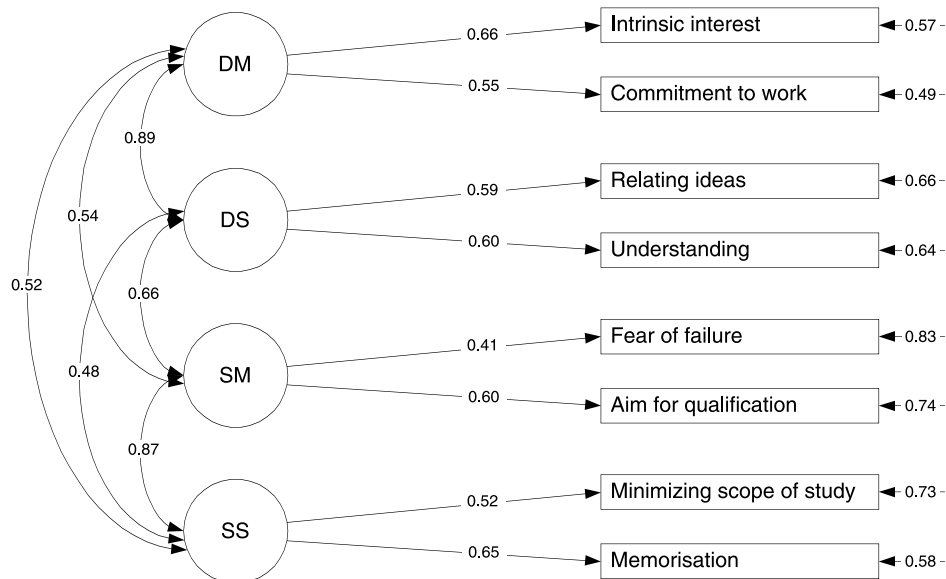


Figure 6. First-order model for motive and strategy scales in LPQ-R-2F.

Note: DM, deep motive; DS, deep strategy; SM, surface motive; SS, surface strategy. All paths are statistically significant at $p < .05$ and $.01$ levels.

The factorial structures concerned, indicated that the LPQ-R-2F and the study approaches could be interpreted hierarchically, as represented by first- and second-order factors. Importantly in this study, as also demonstrated by Kember *et al.* (2004), the two study approaches – deep and surface – were depicted as higher-order factors and were defined by four first-order latent constructs. These four first-order constructs, deep and

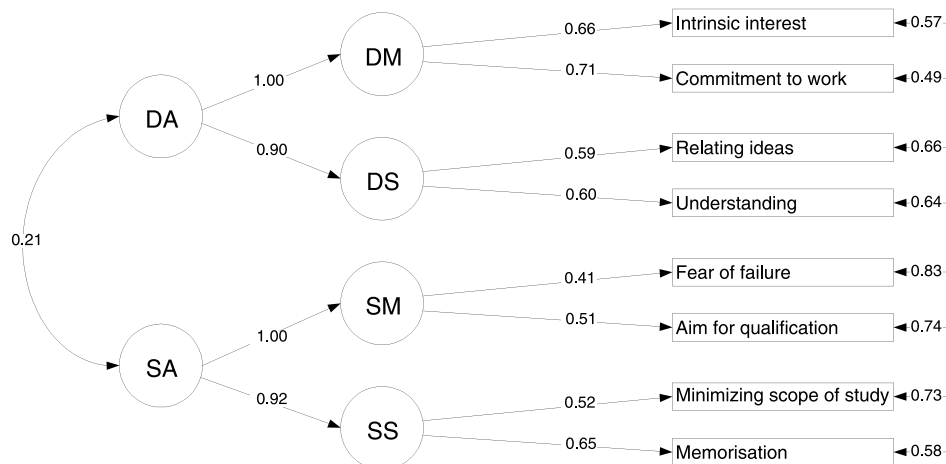


Figure 7. Hierarchical model for motive and strategy scales in LPQ-R-2F.

Note: DM, deep motive; DS, deep strategy; SM, surface motive; SS, surface strategy. All paths are statistically significant at $p < .05$ and $.01$ levels.

surface motives and strategies, were measured by eight corresponding indicators. A full explanation of the structuring of the eight corresponding indicators is presented in the Kember *et al.* study and is not emphasized in this article; however, the evidence established in this study differing from Study I and earlier studies (e.g. Bernardo, 2003; Burnett & Dart, 2000; Cano, 2005a, 2005b; Eklund-Myrskog & Claes-Göran, 1999; Kember & Leung, 1998; Sachs & Gao, 2000) indicates a reconceptualization of the study approaches in terms of their multidimensionality.

GENERAL DISCUSSION

The approaches to study and the LPQ (Biggs, 1987; Marton & Säljö, 1976) have been a recurring theme recently in the literature and have been addressed by a number of researchers (Kember *et al.*, 2004; Sachs & Gao, 2000). Noteworthy, the argument has been advanced from two premises. First, education has undergone a rapid and major transformation since the original conceptualization of the study approaches, and inevitable changes needed to be made (Biggs *et al.*, 2001). Second, contradictory findings which emerged cross-culturally during the 1980s and 1990s indicated a strong need to readdress both the theoretical conception of study approaches and the LPQ. Insights into the study approaches of non-Western students, for example, and a better understanding of extrinsic motivation have contributed to the rethinking of student study approaches and the LPQ's psychometric properties (Kember, Wong, & Leung, 1999).

We reported two studies in this article that investigated the appropriateness of SAL theoretical model (Biggs, 1987) in the South Pacific context using two alternative inventories – the original LPQ and the revised version of the LPQ. Although differing somewhat in their contents and the number of items used, both inventories indicate strong support for two main study approaches and not three, as originally conceptualized by Biggs (1987). The results reported in Study I call into question the achieving approach and advocate instead the two main study approaches – deep and surface. Consistent with earlier findings (Bernardo, 2003; Eklund-Myrskog & Claes-Göran, 1999; Kember & Leung, 1998; Watkins & Akande, 1994; Zhang & Bernardo, 2000), this study validates the conclusion that Richardson (1994) drew, namely:

the broad distinction between two fundamental approaches to studying: first an orientation towards comprehending the meaning of the materials to be learned; and, second, an orientation towards merely being able to reproduce those materials for the purposes of academic assessment. (p. 463)

Empirical results of Study I highlight, importantly, the notion of reading for *understanding* and reading for *reproducing* contents. It is not surprising, given the social, historical and cultural contexts of Fiji (Deo & Phan, 2006; Phan & Deo, 2006, 2007; Tavola, 1991), that secondary students largely engage in this learning orientation. In their articles, Phan and Deo argue that the issue of landlessness, insecurity, discrimination and poverty experienced by the Indo-Fijians, historically and ongoing, serves as a pressure for these students to excel academically. The parental and societal pressure to excel academically, enabling economic and social mobility, instils a mindset for the majority of the Indo-Fijian students enrolled in the regional university of the South Pacific to engage more surface-based and less deep-based orientation in their learning. Phan and Deo suggest that this 'institutionalization' of learning, where students pick up the tricks of effective learning, for example, the usage of 'memorization' and 'rote learning', starts early and is encouraged in schools. For these students, effective learning entails the notion

of being able to pass quizzes, tests and external examinations with good grades and these good grades then allow students to proceed onto the university. The cycle is, in effect, perpetuated at the university wherein students orient towards a reproducing approach so as to achieve the *status quo*.

Analysis of this evidence then, in concert with existing results (e.g. Bernardo, 2003; Eklund-Myrskog & Claes-Göran, 1999; Kember & Leung, 1998; Watkins & Akande, 1994), suggests the importance and rationale for conceptualizing approaches to study as consisting of only two main orientations. The argumentative premise, based on Study I and previous empirical evidence, establishes the need to address alternative inventories other than the original LPQ.

Results obtained from Study II also, in part, support the notion of student learning as being oriented by the desire to learn for meaning and understanding, and that of reproducing for other assessment-related purposes. Equally important is the theoretical validation of student approaches to study by means of a first and second-order factor representation. This evidence accentuates the important theoretical perspective of study approaches (Biggs, 1987; Marton & Säljö, 1976). In itself, as a 'theory-driven' procedure (Byrne, 1998; Mueller, 1996; Pedhazur, 1997), CFA enables the evaluation and subsequent inference and interpretation of hierarchical representations of constructs that is otherwise not possible using alternative statistical means. The psychometric evidence obtained in Study II indicates that study motivation and strategies, corresponding to deep and surface approaches, structure in a complex hierarchical organization that is similar to the way other psychological constructs have been shown to structure (e.g. Bong, 1997; Bong & Clarke, 1999; Bong & Skaalvik, 2003).

The LPQ-R-2F has emerged recently as an alternative inventory (Kember *et al.*, 2004) and more empirical studies are therefore needed to validate the items that make up this inventory. The cultural and social contexts in which the revised LPQ is being administered in Study II need to be acknowledged when discussing generalizations about the psychometric properties of the LPQ-R-2F. For instance, the teaching learning environment in the South Pacific region is defined by a formal teaching style that is largely orientated towards curriculum-based outcomes, rote learning and external examination, especially when compared with the West. The defining characteristics of this schooling system may therefore impact on responses and outcome of the LPQ-R-2F. The schooling system in Hong Kong is very similar to that of the South Pacific region's (Sachs & Gao, 2000), and this may have contributed to the similar findings established in the Kember *et al.* study and this study. Given the marked differences in the teaching learning process between the Western and the non-Western countries, it is important to extend this validation of the revised LPQ with the addition of other Western and non-Western samples. One could argue the possible shortcomings of the existing items, scales and subscales when used in another social and cultural context. There is limited evidence at present, we believe, to warrant the universality of the revised LPQ with different countries and cultures.

Conclusion

Previous research examining student approaches to learning (SAL) with Biggs' (1987) LPQ has reported, in general, a two-factor solution that depicts a reproducing factor and meaning factor. Despite the consistent results of a two-factor solution cross-culturally, there has been raised uncertainties concerning the appropriateness and applicability of the LPQ with different cultural samples (Bernardo, 2003; Zhang & Bernardo, 2000).

This research involving two empirical studies situated in the South Pacific context set out to investigate the conception of study approaches advocated (Biggs, 1987, 1993a, 1993b) using two alternative inventories – the original LPQ and the revised LPQ. The first study primarily focused on examining the theoretical three dimensions of study approaches using the original LPQ. In the second study, the main focus was on the evaluation of the revised LPQ and its psychometric properties.

There are a number of important caveats for future research. First, as discussed earlier, there is a need to continue and extend the reconceptualization of the study approaches to different social and cultural settings. The premise in this examination concerns the appropriateness and relevance of the theoretical conceptions of learning to describe students' study habits in different countries and cultures. Prudent to this premise also, is the challenge for researchers to expand the inquiry concerning the alternative between the original LPQ and the revised LPQ in describing student approaches to learning.

Secondly, the hierarchical representation of study approaches inferring from CFA has also been discussed from a *theoretical* perspective. Researchers could further the investigation by exploring whether the statistically derived higher-order factors, deep and surface approaches, found in Study II reflect students' 'true' experiences of their understanding of different study approaches. One way of examining this research investigation would be to use qualitative and mixed methods paradigms (Creswell, 2003; Tashakkori & Teddlie, 1998). Thirdly, researchers need to expand the *between-construct* validity of the LPQ by testing it with other criterion-based outcomes, such as academic achievement and locus of control.

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